

Hand in

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Problem (1). Let $X = [-1, 1] \times \{-1, +1\} \subseteq \mathbb{R}^2$ have the usual metric topology and define a relation $\sim$ on X saying that $(a, -1) \sim (a, 1)$ for all $a \in [-1, 0) \cup (0, 1]$. Let $Y = X / \sim$ have the quotient topology

- Check that $\sim$ is an equivalence relation (nevermind I guess?)
- Is Y Hausdorff?
- Does Y have a countable basis for its topology?
- Check up the definition of connectedness in section 23: is Y connected?
- Check up the definition of compact in section 26: is Y compact?

Solution and proof of all.

Hausdorff?

Recall that Y is Hausdorff if any two distinct points have some open neighborhoods that are disjoint.

We will use the fact that open sets in X correspond exactly with open sets in Y .

For Y to have the Hausdorff property we would require that, as $(0, 1)$ and $(0, -1)$ are distinct, they have some open disjoint neighborhoods that are disjoint.

Denote $(a, 1) = a_+$ and $(0, -1) = a_-$. Let U be an open neighborhood of 0_+ and V an open neighborhood of 0_- .

Then $q^{-1}(U), q^{-1}(V)$ are open neighborhoods of X . As X has the metric topology we know there are $\varepsilon_+, \varepsilon_- > 0$ such that

$$(-\varepsilon_+, \varepsilon_+) \times \{1\} \subseteq q^{-1}(U)$$

and

$$(-\varepsilon_-, \varepsilon_-) \times \{-1\} \subseteq q^{-1}(V).$$

Choose $0 < a < \min\{\varepsilon_-, \varepsilon_+\}$. Then $a_+ \in q^{-1}(U)$ and $a_- \in q^{-1}(V)$, but

$$a_+ \sim a_-$$

so

$$q(a_+) = q(a_-) \in Y.$$

Call this point y . Then $y \in U \cap V$, meaning they are not disjoint. This violates the Hausdorff property. Intuitively, we will use the fact that some open neighborhood of $(0, 1)$ will have elements whose fibers

Countable basis?

Indeed it does have a countable basis.

X has a countable basis since it is just two copies of $[-1,1] \subseteq \mathbb{R}$ which has a countable basis which we can construct using the rationals.

Let $\mathcal{B} = \{B_1, B_2, \dots\}$ be such a basis. Since q is open ($q^{-1}q(B)$ is open),

$$\mathcal{C} = \{q(B) \mid B \in \mathcal{B}\}$$

consists of open sets in Y . Moreover, if $y \in U$ for some open set in Y , choose $x \in q^{-1}(U)$. Since $q^{-1}(U)$ is open, there is some $B \in \mathcal{B}$ with

$$x \in B \subseteq q^{-1}(U).$$

Therefore $Y \in q(B) \subseteq U$. So $\mathcal{C}$ is a countable basis for Y .

Connected?

The space is indeed connected. I like to use the following procedure.

Let $f : Y \rightarrow \{0,1\}$ be any continuous function where the codomain has the discrete topology.

Let $X_+ = [-1,1] \times \{1\}$ and $X_- = [-1,1] \times \{-1\}$. $f \circ q$ is continuous so the restrictions

$$f \circ q \big|_{X_+} : X_+ \rightarrow \{0,1\}$$

and

$$f \circ q \big|_{X_-} : X_- \rightarrow \{0,1\}$$

are continuous. X_+ and X_- are homeomorphic to $[-1,1]$ and therefore connected so these maps must be constant. For convenience write

$$f \circ q \big|_{X_+} \equiv c_+, \text{ and } f \circ q \big|_{X_-} \equiv c_-$$

where $c_+, c_- \in \{0,1\}$.

Recall that $X_+ \ni a_+ \sim a_- \in X_-$ so $q(a_+) = q(a_-)$. Hence

$$c_+ = f(q(a_+)) = f(q(a_-)) = c_-.$$

As $Y = q(X_+) \cup q(X_-)$ the map f must be constant on the whole domain.

Hence the space is connected!

Compact?

The space is indeed compact.

We know that $[-1,1] \times \{1\}$, $[-1,1] \times \{-1\} \cong [-1,1]$ are compact. So for a covering of X reduce the open covers on each copy of the line to a finite cover.

Now take some open cover $\bigcup_{\alpha} U_{\alpha}$ of Y . Then

$$q^{-1}\left(\bigcup_{\alpha} U_{\alpha}\right) = \bigcup_{\alpha} q^{-1}(U_{\alpha})$$

is a cover of X . Since X is compact we can take a finite number of these

$$\bigcup_{\alpha_i, i \in [n]} q^{-1}(U_\alpha)$$

to cover X . Meaning, every a_+ is contained in some U_{α_i} and every a_- is contained in some U_{α_j} . Then $q(a_+), q(a_-) \in U_{\alpha_i} \cup U_{\alpha_j}$ so the image of this finite cover suffices as a finite covering of Y .

It would also have sufficed to point out that the continuous image of a compact space is compact! $\square$

Problem (2). Let $\mathbb{R}^\omega$ be the countably infinite product of $\mathbb{R}$ with itself. Let $\mathbb{R}^\infty \subset \mathbb{R}^\omega$ be the subset of all sequences $(x_1, x_2, \dots)$ that are eventually zero (i.e., there exists an N such that $x_n = 0$ for all $n > N$).

- a) What is the interior and closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ if $\mathbb{R}^\omega$ is given the product topology? Is $\mathbb{R}^\infty$ (with this topology) Hausdorff?
- b) What is the interior and closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ if $\mathbb{R}^\omega$ is given the box topology? Is $\mathbb{R}^\infty$ (with this topology) Hausdorff?
- c) What is the interior and closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ if $\mathbb{R}^\omega$ is given the uniform topology? Is $\mathbb{R}^\infty$ (with this topology) Hausdorff?

Solution and proof of (a). A basic open neighborhood of $x = (x_1, x_2, \dots)$ looks like

$$\left(\prod_{i \in [N]} U_i \right) \times \mathbb{R} \times \mathbb{R} \times \dots,$$

so only finitely many coordinates are restricted.

For the interior, take any $x \in \mathbb{R}^\infty$. Any neighborhood of x leaves all sufficiently large coordinates completely unrestricted. Hence inside that neighborhood we can find a sequence y with, e.g. $y_n = 1$ for infinitely many unrestricted coordinates.

Then y is not eventually zero. Therefore no open neighborhood of x lies entirely inside $\mathbb{R}^\infty$. Hence the interior $\text{Int}(\mathbb{R}^\infty) = \emptyset$.

For closure, take an arbitrary $x = (x_1, x_2, \dots) \in \mathbb{R}^\omega$. Take a neighborhood restricting only the first N coordinates. Choose $y = (x_1, \dots, x_N, 0, 0, \dots)$. Then

$$y \in \mathbb{R}^\infty$$

and y lies in the given neighborhood of x . Thus every open neighborhood of every $x \in \mathbb{R}^\omega$ intersects $\mathbb{R}^\infty$. So $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$, meaning $\mathbb{R}^\infty$ is dense in $\mathbb{R}^\omega$ under the product topology.

The space is Hausdorff since distinct sequences differ in at least one coordinate and $\mathbb{R}$, as well as subspaces of $\mathbb{R}$, are Hausdorff. $\square$

Solution and proof of (b). Now a neighborhood is much smaller

$$\prod_{n=1}^{\infty} U_n,$$

where every $U_n \subseteq \mathbb{R}$ is open.

Suppose $x \in \mathbb{R}^{\infty}$, so for some N , $x_n = 0$ ($n > N$).

Take any box neighborhood

$$\prod_{n=1}^{\infty} U_n$$

of x . For every $n > N$, $0 \in U_n$. Since U_n is an open neighborhood of 0, it contains some nonzero y_n . Then the resulting sequence lies in the box neighborhood, but is not eventually zero. Therefore the interior is empty.

Now suppose $x \notin \mathbb{R}^{\infty}$. Then $x_n \neq 0$ for infinitely many n . Let $I = \{n \mid x_n \neq 0\}$.

For every $n \in I$, choose an open interval U_n around x_n such that $0 \notin U_n$. For $n \notin I$ just take $U_n = \mathbb{R}$. Then

$$U = \prod_{n=1}^{\infty} U_n$$

is an open neighborhood of x .

Every $y \in U$ satisfies $y_n \neq 0$ for every $n \in I$, and I is infinite. Hence no $y \in U$ is eventually zero.

So

$$U \cap \mathbb{R}^{\infty} = \emptyset.$$

Therefore every point outside $\mathbb{R}^{\infty}$ has a neighborhood avoiding $\mathbb{R}^{\infty}$. Thus $\overline{\mathbb{R}^{\infty}} = \mathbb{R}^{\infty}$. So in the box topology it is closed, but still has empty interior.

The space is Hausdorff for the same reason as in (a). □

Solution and proof of (c). For $0 < \varepsilon < 1$ and ε -ball around x is

$$B_{\varepsilon}(x) = \{y = (y_n) \in \mathbb{R}^{\omega} \mid |x_n - y_n| < \varepsilon \text{ for every } n\}.$$

Take any $x \in \mathbb{R}^{\infty}$. Then there is sufficiently large N so that $x_n = 0$ whenever $n \geq N$. Let $\varepsilon > 0$. We want to show $B_{\varepsilon}(x)$ contains some point not in $\mathbb{R}^{\infty}$. Define y_n as

$$y_n = \begin{cases} x_n, & n \leq N, \\ \varepsilon/2, & \text{otherwise.} \end{cases}$$

Then for every n ,

$$|x_n - y_n| < \varepsilon,$$

so $y \in B_\varepsilon(x)$, but $y_n = \varepsilon/2 > 0$ for $n > N$ so $y \notin \mathbb{R}^\infty$. Therefore the interior is empty.

We claim the closure is

$$\overline{\mathbb{R}^\infty} = \{x = (x_n) \mid x_n \rightarrow 0\}.$$

Call the set on the right Ω .

If $x_n \rightarrow 0$ then x is in the closure. Take any $\varepsilon > 0$. Since $x_n \rightarrow 0$ there is N such that $|x_n| < \varepsilon$ for $n > N$. Truncate x to

$$y = (x_1, \dots, x_N, 0, 0, \dots).$$

Then $y \in \mathbb{R}^\infty$. Also

$$|x_n - y_n| = |x_n| < \varepsilon, \quad n > N$$

and 0 otherwise. Therefore $y \in B_\varepsilon(x)$. Therefore every uniform neighborhood of x intersects $\mathbb{R}^\infty$ so $x \in \overline{\mathbb{R}^\infty}$.

We conclude $\Omega \subseteq \overline{\mathbb{R}^\infty}$.

Now suppose $x_n \not\rightarrow 0$. Then there is some $\varepsilon > 0$ so that $|x_n| \geq \varepsilon$. Consider $B_{\varepsilon/2}(x)$. Suppose $y \in \mathbb{R}^\infty$. Then there is N so that $y_n = 0$ for $n > N$, but there are infinitely many n for which $|x_n| \geq \varepsilon > \varepsilon/2$ so

$$y_n \notin B_{\varepsilon/2}(x).$$

Then $B_{\varepsilon/2} \cap \mathbb{R}^\infty = \emptyset$ so $x \notin \overline{\mathbb{R}^\infty}$. Therefore

$$\overline{\mathbb{R}^\infty} = \Omega.$$

$\mathbb{R}^\omega$ is Hausdorff since the uniform topology is induced by a metric. The subspace $\mathbb{R}^\infty$ is therefore Hausdorff. $\square$

Problem (3). Let X be a topological space, Y a set and let $f: X \rightarrow Y$ be a surjection.

Show that the quotient topology on Y is the unique topology so that

if Z is a topological space and $g: Y \rightarrow Z$ is a function then g is continuous if and only if gf is continuous.

Proof. With $f: X \rightarrow Y$ some surjection, the quotient topology on Y is the collection

$$\tau_f = \{U \subseteq Y \mid f^{-1}(U) \text{ is open in } X\}.$$

Equivalently, U is open in Y if and only if its preimage under f is open in X .

We first prove existence. Assume Y has the quotient topology and let $g : Y \rightarrow Z$.

$$g \text{ continuous} \Rightarrow g \circ f \text{ continuous}$$

is immediate as f is continuous and the composition of continuous functions is continuous.

For the converse, suppose the composition $g \circ f$ is continuous. Then for an open set U in Z the preimage

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U)) \subseteq X$$

is open. Then, by assumption, $g^{-1}(U) \subseteq Y$ is open. So the preimage of open sets under g is open, hence it is continuous.

For uniqueness, suppose τ is some other topology on Y having the property.

Let $Z = (Y, \tau_f)$ and $g = \text{id}_Y : (Y, \tau) \rightarrow (Y, \tau_f)$. Then

$$g \circ f = f : X \rightarrow (Y, \tau_f)$$

which is continuous.

Since τ has the stated property, g is therefore continuous. Hence every τ_f -open set is τ -open, so

$$\tau_f \subseteq \tau.$$

For the opposite inclusion, observe first that

$$f : X \rightarrow (Y, \tau)$$

is continuous. Indeed, taking

$$g = \text{id}_Y : (Y, \tau) \rightarrow (Y, \tau)$$

in the assumed property, g is continuous, and hence $g \circ f = f$ is continuous. Now consider

$$h = \text{id}_Y : (Y, \tau_f) \rightarrow (Y, \tau).$$

Since

$$h \circ f = f : X \rightarrow (Y, \tau)$$

is continuous, the universal property already proved for the quotient topology implies that h is continuous. Therefore every τ -open set is τ_f -open, and hence

$$\tau \subseteq \tau_f.$$

Consequently,

$$\tau = \tau_f,$$

so the quotient topology is unique. □

Problem (Munkres 17.12). Show that a subspace of a Hausdorff space is Hausdorff.

Proof. Let X be a Hausdorff space and let $A \subseteq X$ be a subspace, i.e. having the topology

$$\tau_A = \{U \cap A \mid U \subseteq X \text{ open}\}.$$

Take any two distinct $x, y \in A$. Then there exist open $U_x, U_y \subseteq X$ such that $x \in U_x$, $y \in U_y$ and $U_x \cap U_y = \emptyset$.

Then

$$A \cap U_x \subseteq U_x$$

and

$$A \cap U_y \subseteq U_y$$

with

$$(A \cap U_x) \cap (A \cap U_y) = A \cap (U_x \cap U_y) = A \cap \emptyset = \emptyset.$$

Thus we have found disjoint open sets in A $A \cap U_x$, $A \cap U_y$ containing x and y , respectively.

As x, y were arbitrary, the Hausdorff property is satisfied. $\square$

Problem (Munkres 17.13). Show that X is Hausdorff if and only if the diagonal $\Delta = \{x \times x \mid x \in X\}$ is closed in $X \times X$.

Proof. Suppose X is Hausdorff. Then for any $x, y \in X$ we can find disjoint open neighborhoods U_x, U_y for them.

Take $(x, y) \in X \times X \setminus \Delta$. Since $x \neq y$ there are disjoint open sets U_x, U_y for the points. Then $U_x \times U_y$ cannot contain a point of the form (z, z) . Suppose $(z, z) \in U_x \times U_y$. Then $z \in U_x$ and $z \in U_y$ so $U_x \cap U_y \neq \emptyset$ which is a contradiction. Thus every point outside Δ has an open neighborhood disjoint from Δ so $X \times X \setminus \Delta$ is open and hence Δ is closed.

Now suppose Δ is closed, meaning $X \times X \setminus \Delta$ is open. Take distinct $x, y \in X$. Then $(x, y) \notin \Delta$ so there is an open set $U \times V \subseteq X \times X \setminus \Delta$ containing (x, y) .

The claim then is that $U \cap V = \emptyset$. Suppose not. If some $z \in U \cap V$, then $(z, z) \in U \times V$, but $(z, z) \in \Delta$ so we arrive at a contradiction. Since U, V are open neighborhoods of x and y , respectively and those points were arbitrary, the space is Hausdorff. $\square$

Problem (Munkres 20.3). Let X be a metric space with metric d .

a) Show that $d : X \times X \rightarrow \mathbb{R}$ is continuous.

b) Let X' denote a space having the same underlying set as X . Show that if $d : X' \times X' \rightarrow \mathbb{R}$ is continuous, then the topology of X' is finer than the topology of X .

Proof of (a). The metric d satisfies

$$\begin{aligned} d(x, y) &\geq 0 \\ d(x, y) &= 0 \Leftrightarrow x = y \\ d(x, y) &= d(y, x) \\ d(x, z) &\leq d(x, y) + d(y, z). \end{aligned}$$

In particular,

$$|d(x, y) - d(x', y')| \leq d(x, x') + d(y, y').$$

Fix some $(x_0, y_0) \in X \times X$. Let $\varepsilon > 0$ and consider the product neighborhood

$$B_{\varepsilon/2}(x_0) \times B_{\varepsilon/2}(y_0).$$

If $(x, y) \in B_{\varepsilon/2}(x_0) \times B_{\varepsilon/2}(y_0)$, then

$$d(x, x_0) < \frac{\varepsilon}{2}, \quad d(y, y_0) < \frac{\varepsilon}{2}.$$

By the triangle-inequality,

$$d(x, y) \leq d(x, x_0) + d(x_0, y_0) + d(y_0, y),$$

so

$$d(x, y) - d(x_0, y_0) \leq d(x, x_0) + d(y, y_0).$$

Therefore

$$|d(x, y) - d(x_0, y_0)| \leq d(x, x_0) + d(y, y_0) < \varepsilon.$$

Hence

$$d(x, y) \in d(x_0, y_0) \pm \varepsilon$$

which is open in $\mathbb{R}$. □

Proof of (b). Write $X = (S, \tau_d)$ where τ_d is the induced topology from d and $X' = (S, \tau')$ be the same underlying set S , but with some other topology τ' . We assume $d : X' \times X' \rightarrow \mathbb{R}$ is continuous.

Fix $a \in X$ and $\varepsilon > 0$. Consider the map $i_a : X' \rightarrow X' \times X'$ by $i_a(x) = (x, a)$. The map is continuous and by assumption $d : X' \times X' \rightarrow \mathbb{R}$ is continuous so

$$d \circ i_a : X' \rightarrow \mathbb{R}$$

is aswell. Then

$$(d \circ i_a)(x) = d(x, a).$$

Consider now $(\infty, \varepsilon) \subseteq \mathbb{R}$. The preimage under $(d \circ i_a)$ is $B_d(a, \varepsilon) = \{x \in X \mid d(x, a) < \varepsilon\}$. Thus the metric ball $B_d(a, \varepsilon)$ is open in X' .

Now let U be any open set in X . Since X has the metric topology, for all $x \in U$ there exists $\varepsilon_x > 0$ such that

$$B_d(x, \varepsilon_x) \subseteq U.$$

Hence

$$U = \bigcup_{x \in U} B_d(x, \varepsilon_x).$$

Each ball is open in X' , so U is open in X . Therefore $\tau_d \subseteq \tau'$. $\square$